Network and Emotional Structure of Text as a Cognitive Signal

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An exploratory analysis of LLM-simulated Alzheimer's and healthy-control text using Textual Forma Mentis Networks

Abstract

This exploratory report analyzes textual differences between synthetic datasets resembling Alzheimer's disease (AD) and healthy control (HC) writing. Text was modeled through the framework of Textual Forma Mentis Networks (TFMNs) at both the syntactic and emotional level, revealing a clear structural and emotional distinction between the two simulated groups. A classification analysis on emotion features reached unusually high accuracy, which a follow-up robustness check (LDA/PCA separability plus surface-level sanity checks on the raw text) traces to the internal homogeneity of LLM-generated text rather than to a validated clinical marker.

1. Introduction

Dementia leaves a measurable trace in language. As the disease progresses, patients tend to produce shorter, more repetitive, and less structurally complex sentences, reflecting the gradual deterioration of the semantic memory system that underlies word retrieval and concept association (Huff, Corkin, & Growdon, 1986). Standard NLP methods such as TF-IDF capture some of this through word frequency, but treat words as independent tokens rather than as a structured web of associations.

This project instead models each text as a network, where words are nodes and syntactic relationships are edges. This representation — a Textual Forma Mentis Network (TFMN) — has previously been used to study public perception (Stella, 2020), creativity (Haim et al., 2026), and authorship attribution (Segarra, Eisen, & Ribeiro, 2013). Here it is applied to a two-group comparison: text simulating Alzheimer's disease patients versus age-matched healthy controls.

Two hypotheses guided the analysis: (1) AD-simulated text shows a degraded syntactic network — smaller, less connected, more reliant on repeated concepts than HC-simulated text; and (2) AD-simulated text shows reduced emotional expressivity, consistent with the affective flattening reported in the clinical literature. A third, more exploratory question was whether the resulting features could classify the two groups — and, if accuracy came out unexpectedly high, whether that reflected a real signal or an artifact of how the data was generated.

2. Data

The dataset is entirely synthetic. Text was generated with a large language model (Mistral Large 3), prompted to describe a typical day at a local market while writing in a style characteristic of either an AD patient or a healthy control — incorporating the semantic, syntactic, and emotional patterns associated with each profile.

Using LLM-generated text instead of real patient transcripts removes any privacy concern and allows generating data at scale, but carries a clear risk: synthetic text can end up too internally consistent within each group and too cleanly different between groups, in a way that doesn't reflect how real patients and controls actually write. That risk turns out to matter for how the classification results should be read (see Results and Discussion).

3. Method

3.1 Network construction

Each text was parsed with spaCy's dependency parser and converted into a TFMN via EmoAtlas, linking words based on syntactic proximity in the dependency tree rather than simple co-occurrence. From each network, 11 topological features were extracted: number of nodes and edges, density, mean degree, average clustering coefficient, number of connected components, number of isolated nodes, and — restricted to the largest connected component — its size, size ratio, average shortest path length, and diameter.

3.2 Emotional profiling

EmoAtlas computed emotion z-scores for each text by cross-referencing tokens against the NRC Emotion Lexicon, mapped onto Plutchik's eight core emotions (joy, trust, fear, surprise, sadness, disgust, anger, anticipation). Unlike a transformer-based sentiment model, this approach combines syntactic parsing with a psychologically grounded lexicon, keeping the emotion scores interpretable and computationally light.

3.3 Statistical comparison

Each topological feature was compared between groups with a two-sided Mann-Whitney U test, chosen given the non-normal, bounded nature of network metrics, with Benjamini-Hochberg FDR correction applied across all 11 comparisons.

3.4 Classification and robustness checks

Logistic regression and a decision tree were trained on the 8 emotion z-scores under 10-fold stratified cross-validation, evaluated on accuracy and ROC AUC. Emotion features were chosen over topological ones specifically to reduce dependence on document length, a known confound with LLM-generated text of varying length. Three additional classifiers (random forest, gradient boosting, XGBoost) and the full topological+emotional feature set were tested as an extended comparison, alongside train-test AUC gap as a first overfitting check.

Given the risk of overly homogeneous synthetic data, a Linear Discriminant Analysis and a Principal Component Analysis were run on the same 8 emotion features to check whether the two groups separated cleanly even along an unsupervised axis. As a further, more direct check, four sanity analyses were run on the raw text: (1) TF-IDF plus logistic regression directly on the raw text, (2) inspection of which individual tokens drive that separation, (3) basic stylometric features (document length, lexical diversity, filler words), and (4) classification using document length alone as the only feature.

4. Results

4.1 Network structure

All 11 topological features differed significantly between groups after FDR correction (all adjusted $p < .05$). AD-simulated networks were consistently smaller: fewer nodes (41.6 vs. 87.5), fewer edges (108.3 vs. 262.5), and a smaller largest connected component (38.1 vs. 84.6). They also had shorter average path lengths (2.49 vs. 3.93) and smaller diameters (5.24 vs. 9.01) — smaller and structurally simpler networks, not just smaller ones. AD networks were also more fragmented, with more connected components (2.58 vs. 1.25) and more isolated nodes (0.69 vs. 0.03), suggesting disconnected conceptual islands not integrated into the main discourse. One counter-intuitive result: AD networks were denser (0.13 vs. 0.07) — but density scales inversely with network size, so this very likely reflects fewer nodes rather than richer connectivity.

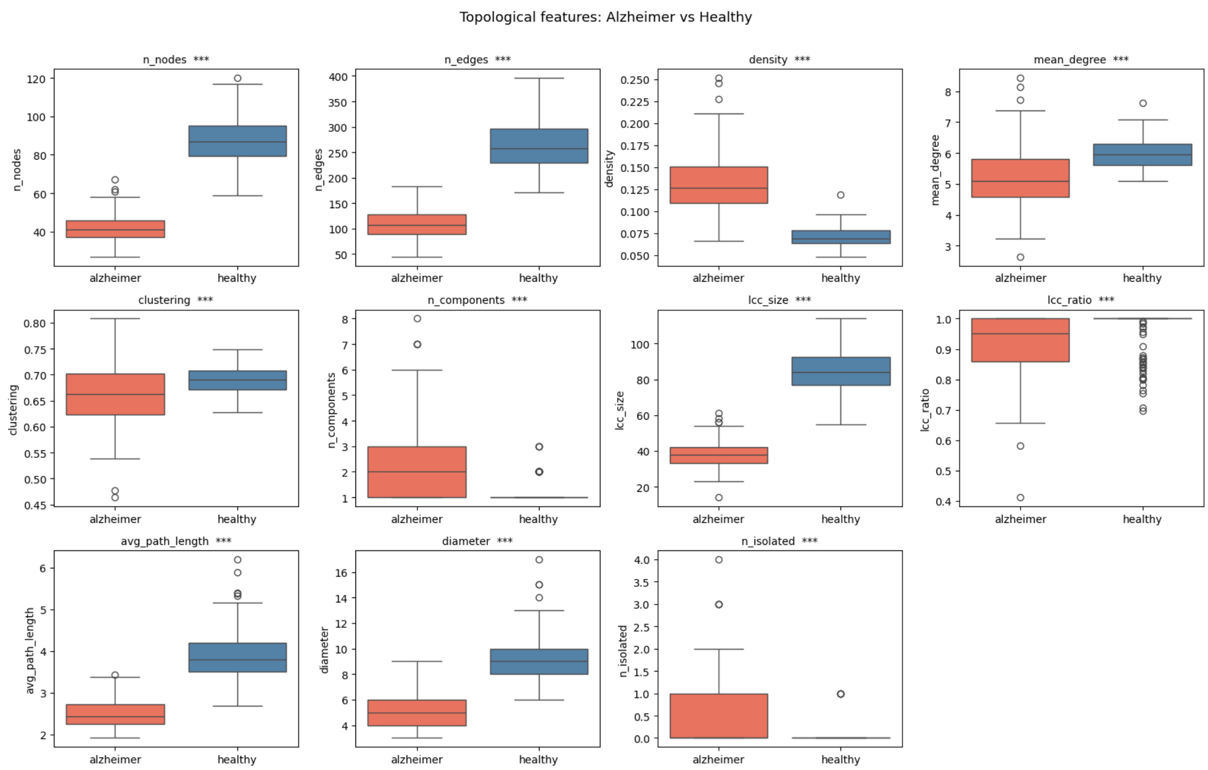

*Figure 1. Topological features compared between AD-simulated and healthy-simulated text. All differences are significant after FDR correction (*** $p < .001$).*

4.2 Emotional profile

Healthy-simulated text showed a clearly more differentiated emotional signature: joy, anticipation, trust, and surprise all stood out well above baseline, while fear, anger, sadness, and disgust fell well below it. AD-simulated text followed the same overall direction but in an attenuated form — trust and disgust, in particular, were statistically indistinguishable from a random baseline. This pattern is consistent with the flattened affect reported in the clinical literature on Alzheimer's disease.

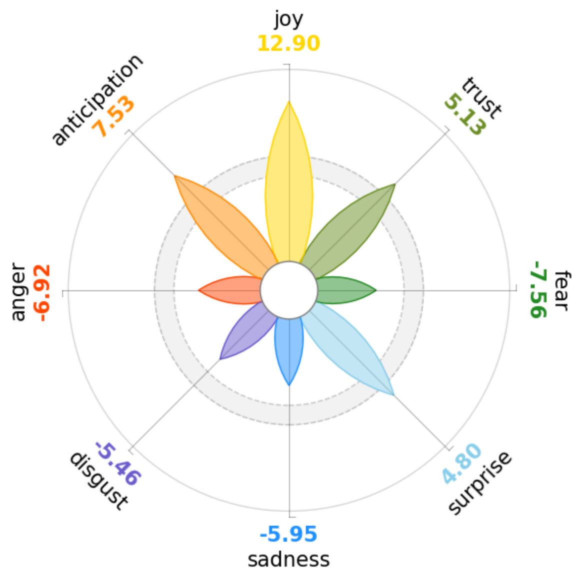

Figure 2. Emotion z-scores for the healthy-simulated corpus, mapped onto Plutchik's wheel. Positive emotions dominate well above baseline.

The aggregated word-association networks for each group illustrate the same pattern qualitatively: the healthy-simulated network centers on rich, sensory, descriptive vocabulary, while the AD-simulated network centers on repetition and word-finding difficulty.

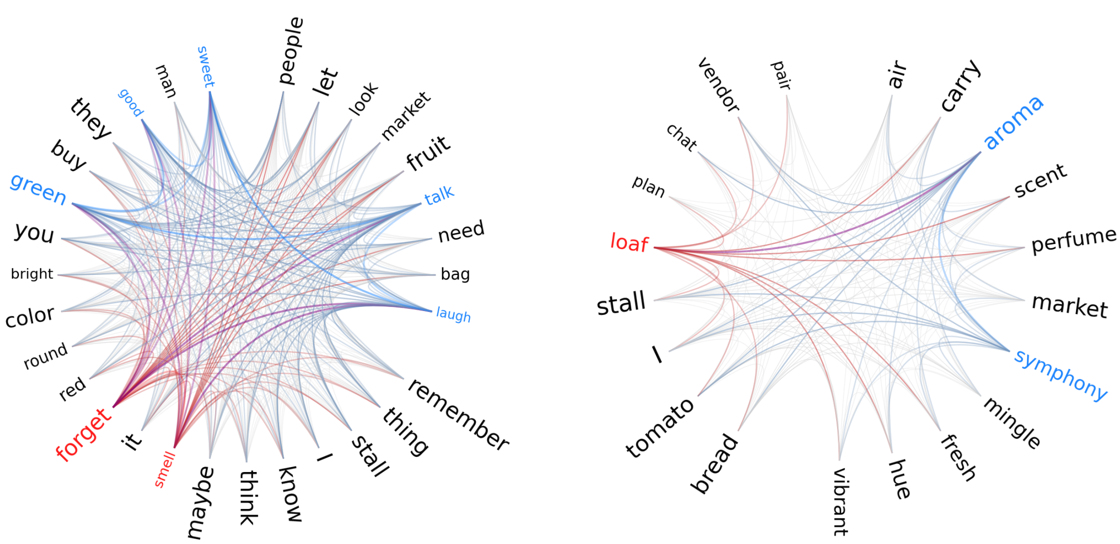

Figure 3. Word-association networks (top-degree words) for the AD-simulated corpus (left) and healthy-simulated corpus (right).

4.3 Classification and predictability

Both classifiers performed strikingly well on emotion features alone: logistic regression reached AUC = 0.97 and accuracy = 0.91; the decision tree reached AUC = 0.93 and accuracy = 0.87. That level of performance from just 8 features on a two-class problem is unusually high, so it was checked rather than taken at face value.

Classifier	AUC	Accuracy
Logistic Regression	0.97	0.91
Decision Tree	0.93	0.87

Both LDA and PCA showed a near-total, one-dimensional separation between the two groups — meaning the groups were separable even without using the labels at all.

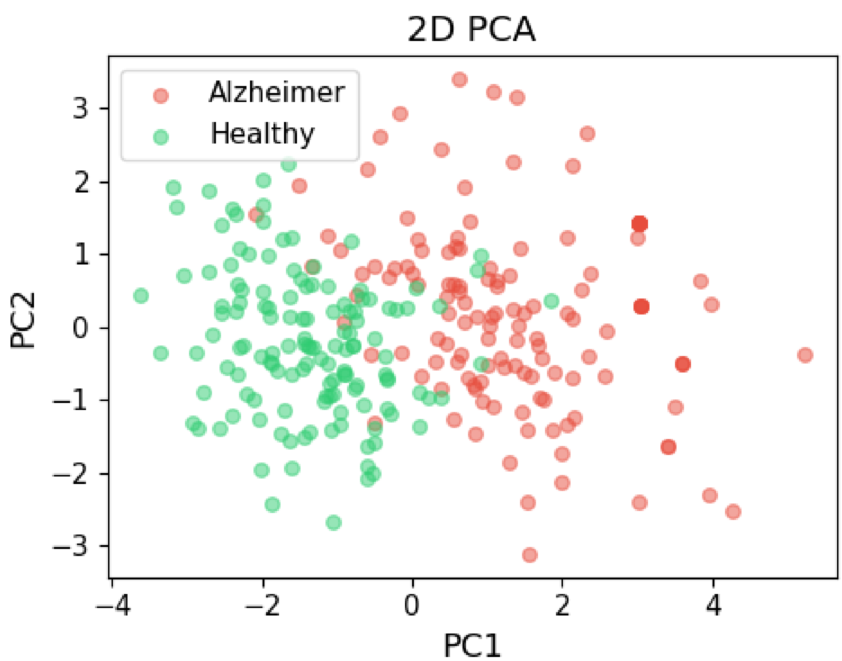

Figure 4. 2D PCA of the 8 emotion features. Groups separate almost completely without using the class labels.

SHAP analysis on the full feature set showed network-size features (number of edges, number of nodes, largest-component size, average path length) dominating the model's decisions far more than any individual emotion — a first sign the classifier may be leaning on document size rather than emotional content.

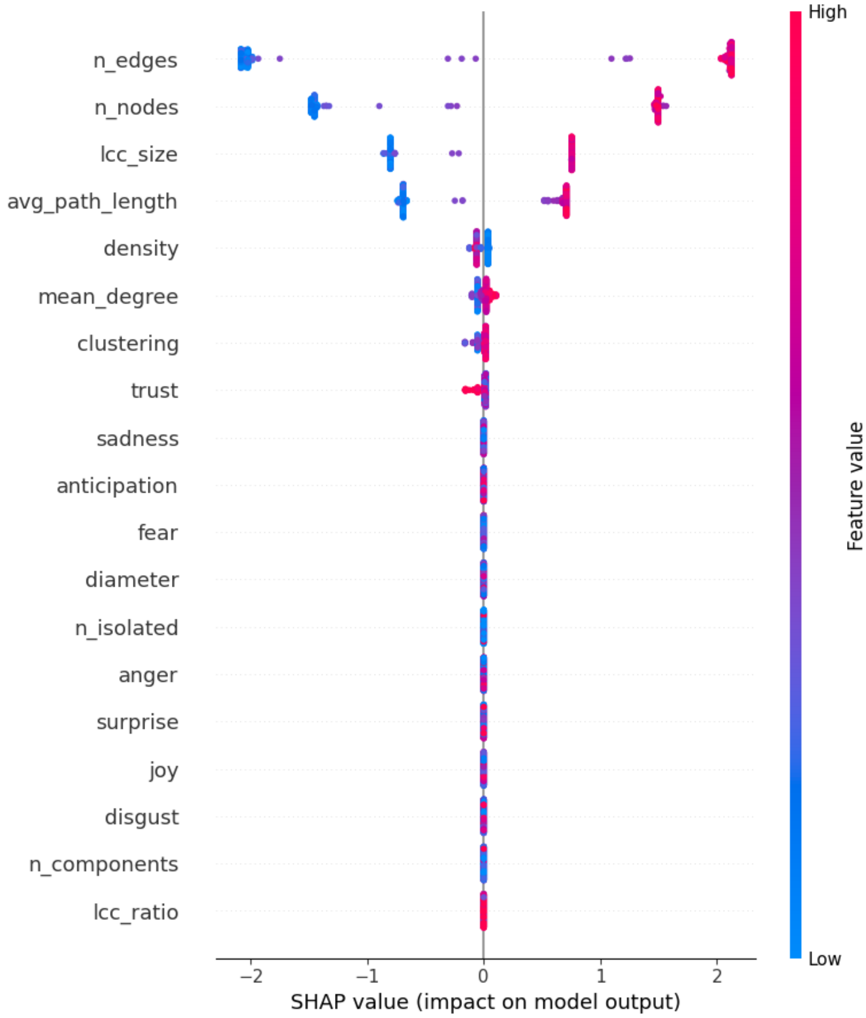

Figure 5. SHAP summary, full topological + emotional feature set. Network-size metrics dominate feature importance.

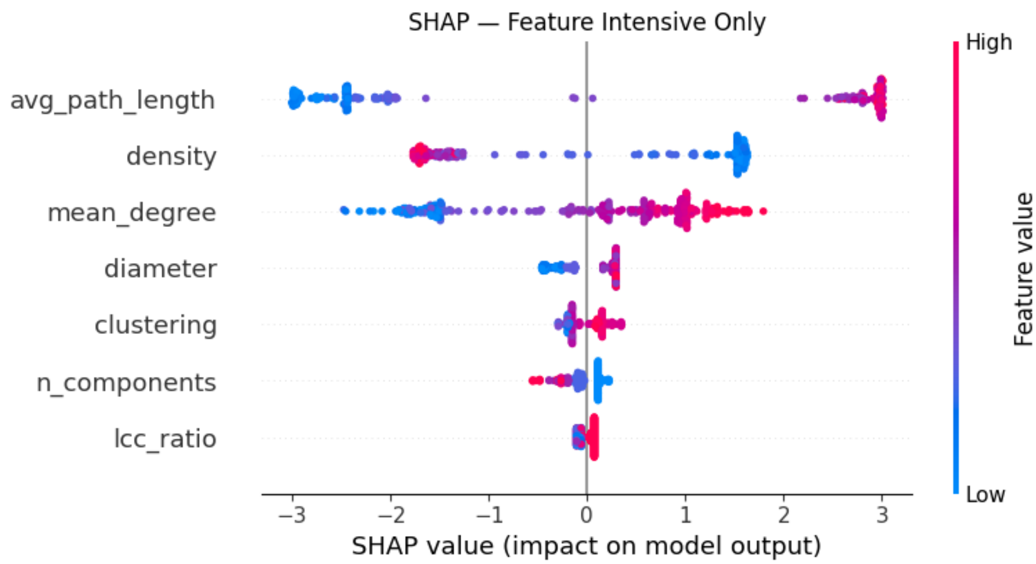

Figure 6. SHAP summary restricted to length-independent (intensive) topological features. Average path length, density, and mean degree still separate the classes, though not fully free of a relationship with text size.

Direct sanity checks on the raw text confirmed the concern raised by the LDA/PCA result. TF-IDF features computed directly on the unprocessed text, with no network or emotion modeling at all, reached an AUC comparable to the engineered-feature classifiers. Inspecting the individual tokens driving that separation showed generic AD-narrative markers (e.g. repetition, hedging language) on one side and vivid, sensory vocabulary on the other — direct evidence that the language model encoded the class distinction into surface word choice. Document length alone, used as the only classification feature, also separated the groups well above chance. Taken together, these checks indicate that a meaningful share of the classification signal is attributable to surface-level stylistic differences in how the LLM generated each condition, rather than to a subtle, emotionally or structurally grounded clinical marker.

5. Discussion

The topological results line up well with what's known about semantic-syntactic decline in dementia (Liu et al., 2022): smaller networks, shorter paths, more fragmentation. The emotional results also track the clinical picture of anhedonia and affective flattening reported in Alzheimer's disease (Shaw et al., 2021).

The classification result is where this analysis earns its exploratory label. High accuracy and AUC would normally be good news, but the LDA/PCA check, the SHAP feature-dominance pattern, and the direct text-level sanity checks all point the same direction: this particular synthetic corpus is probably too internally homogeneous, and too cleanly different between groups, for the classification number to be read as a validated clinical signal. Emotion features were deliberately chosen over topological ones to reduce sensitivity to surface-level text length, on the reasoning that emotional tone is less tied to how much a model writes — the results here show that reasoning wasn't sufficient on its own: emotion features weren't fully immune to the same homogeneity confound.

This isn't a reason to discard the approach. It's a reason to treat the classification number as a property of this particular synthetic dataset, not as evidence that TFMN plus emotion features can detect

Alzheimer's disease from text in general. The natural next step is running the same pipeline on real transcript data — the DementiaBank Pitt Corpus or the ADReSS challenge dataset are the standard benchmarks — and comparing classification performance on synthetic versus real data directly. If the same features that separate LLM-generated groups almost perfectly turn out to separate real patients and controls only modestly, that gap is itself an informative and reportable finding.

6. Conclusion

Modeling text as a network surfaces structural differences between simulated Alzheimer's and healthy writing that align with the clinical literature on semantic-syntactic decline, and the emotional analysis picks up a plausible signature of affective flattening. The classification result is a good example of why a good result deserves the same scrutiny as a bad one: near-perfect separability on synthetic text is more likely to reflect how the data was generated than a validated clinical marker, and the LDA/PCA and sanity checks are what catch that before the number gets over-interpreted.

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